

SPIRE/SPIFFE: Concepts, Configuration, and Infrastructure

What Is Running on This Cluster

Trust domain: `ipc.local`

SPIRE Server — StatefulSet (1 replica) on the cluster. CA keys and registration entries live in SQLite on a 1Gi PVC provisioned by the `nfs-subdir-external-provisioner` StorageClass, backed by nazgul (192.168.89.2) at `/mnt/primary_storage/k8s-nfs`. Listens on gRPC port 8081 via a headless service. Flux manages it from `manifests/spire/`.

SPIRE Agent — DaemonSet on all 6 nodes.

`hostPID: true` is required for workload attestation. When a workload calls the Workload API socket, the agent reads `SO_PEERCRED` from the Unix domain socket connection, which gives it the calling process's PID. Under normal Kubernetes pod isolation, PIDs are namespaced — each pod sees its own PID 1 and the host processes are invisible. With `hostPID: true` the agent's container shares the host PID namespace, so `SO_PEERCRED` returns the *host* PID of the calling process. The agent then reads `/proc/<host-pid>/cgroup` to extract the container ID, and from there calls the kubelet API to get the pod's namespace, service account, and labels. Without `hostPID: true`, `SO_PEERCRED` returns PID 0 and the entire workload attestation chain breaks.

The Workload API socket at `/run/spire/sockets/agent.sock` is a Unix domain socket created by the agent inside the container. The DaemonSet mounts `/run/spire/sockets` as a `hostPath` volume with `DirectoryOrCreate`, which means the directory is created on the node's filesystem if it doesn't exist, and the socket file lives there rather than inside the container's ephemeral overlay filesystem. Any pod on the same node that mounts `/run/spire/sockets` as a `hostPath` gets access to the same socket — this is how workload containers reach the agent without any service discovery or network routing. The socket is node-local by design: the agent on `ipc7` only knows about workloads on `ipc7`, and those workloads only connect to `ipc7`'s agent socket.

Each agent bootstraps its trust bundle via the `verify-bundle` init container (see TPM Server Attestation below), which fetches the bundle signed by the server's TPM key, verifies the signature, and writes the bundle to an `emptyDir` volume shared between the init container and the main container. Init containers and main containers never run simultaneously — Kubernetes enforces that all init containers exit before the main containers start — so the `emptyDir` is the handoff point: the init container writes the verified bundle and exits, then the agent starts and reads it. The server also maintains the `spire-bundle` ConfigMap via the `k8sbundle` notifier, but agents no longer bootstrap from it directly.

Cluster Architecture

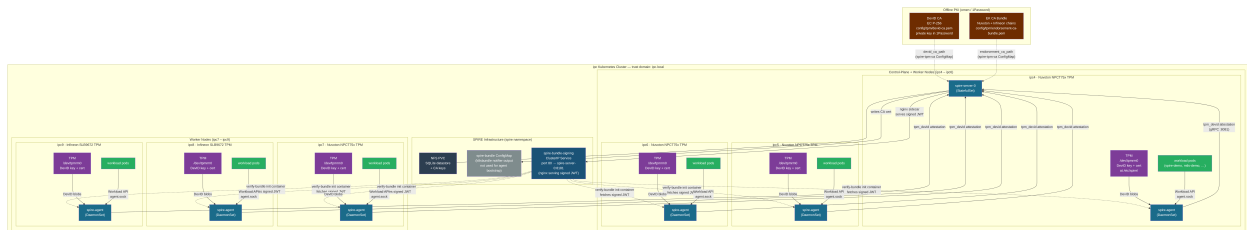


Figure 1: SPIRE cluster architecture — nodes, agents, server, TPMs, and workloads

SPIFFE and SPIRE — the Concepts

SPIFFE (Secure Production Identity Framework For Everyone) is a standard, not software. It defines:

- A URI format for workload identity: `spiffe://<trust-domain>/<path>`
- What a valid credential (SVID) looks like
- How an agent presents that credential to workloads (the Workload API)

SPIRE is the reference implementation of SPIFFE: a server that acts as a CA and a registration database, plus per-node agents that handle attestation and SVID issuance.

What Is an SVID, and Who Is Attesting to Whom?

An SVID is an **identity document**, not a safety certificate. The analogy is a passport: a passport doesn't say "this person is trustworthy" — it says "this person is who they claim to be, and a government vouches for that." What the receiving party does with that identity is their own business.

This distinction matters because attestation flows in multiple directions simultaneously. Here is the full chain as implemented on this cluster:

0. Agent trusts the server (TPM-signed bundle) Before the SPIRE agent starts, the `verify-bundle` init container fetches the trust bundle JWT from `spire-bundle-signing` — a ClusterIP Service that exposes an HTTP endpoint on the SPIRE server pod (a busybox httpd sidecar on port 8181, serving the file that the `signer-unix` sidecar writes after signing with the server's TPM key). The init container verifies the RS256 signature on that JWT against `/etc/spire/server-bundle-signing.pub` and only writes the bundle to disk if the signature is valid. The agent will not start if this check fails.

The server's TPM public key lives in a file on the agent node rather than being fetched from the TPM at runtime for two reasons. First, TPMs are local hardware — the server's TPM is physically on ipc4 and there is no network protocol for reading a remote TPM's public key; each TPM is only accessible to the machine it is soldered to. Second, and more importantly, fetching the key through any Kubernetes channel (ConfigMap, Secret, API) would defeat the purpose entirely: the whole point is that the agent needs a trust anchor that is independent of Kubernetes infrastructure, so that a compromised control plane cannot substitute a rogue server's key. A key that arrives via Kubernetes can be swapped by anyone who controls Kubernetes.

Writing the key to `/etc/spire/server-bundle-signing.pub` during physical provisioning (via `provision-tpm-devid.sh`, running from omen over SSH before SPIRE starts) is what makes it out-of-band. It arrives through the same direct-SSH channel as the DevID material — a channel that SPIRE cannot influence and that a Kubernetes compromise cannot reach. This means the agent's trust in the server is rooted in the server's hardware TPM, not in Kubernetes infrastructure.

1. Server trusts the agent (TPM DevID credential activation) DevID (IEEE 802.1AR Secure Device Identity) is the standard that defines a hardware-bound device credential: a key pair generated inside the TPM by the local administrator during enrollment (not at manufacturing — it is specific to this cluster's identity scheme, not indelible), so the private key never leaves the chip, paired with a certificate signed by an authority that vouches for which device the key belongs to. The agent sends its DevID certificate (signed by the cluster DevID CA) and proves it holds the

corresponding private key by signing a server-issued nonce. The server then performs a credential activation challenge using the node's EK (Endorsement Key) public key. The EK is a key pair burned into the TPM at manufacturing time; the manufacturer signs an EK certificate and stores it in the TPM's non-volatile memory, making the EK the hardware root of trust for that physical chip. The credential activation challenge encrypts a secret to the EK public key such that only the TPM holding that EK can decrypt it, and only if the DevID key is also loaded in that same TPM at the same time. The agent's TPM decrypts and returns the secret. Both proofs together — key possession and TPM residency — satisfy the server that the agent is running on a real, enrolled node.

2. Agent attests the workload (SO_PEERCRED + kubelet) When a workload process connects to `agent.sock`, the agent reads `SO_PEERCRED` from the Unix socket to get the caller's host PID, walks `/proc/<pid>/cgroup` to extract the container ID, and calls the kubelet API to get the pod's namespace, service account, and labels. The workload never claims its own identity — the agent determines it by observation, from outside the workload.

3. Server issues an SVID to the workload via the agent The agent presents the observed workload identity to the server. If it matches a registration entry, the server signs an X.509 SVID and returns it to the agent, which hands it to the waiting workload process.

4. Workload presents the SVID to a peer The workload now has a cryptographically-signed identity document. When it connects to another service — `mtls-server`, Vault, an API gateway — it presents the SVID as its TLS client certificate.

5. Peer verifies against the trust bundle The receiving service verifies the SVID's signature against the SPIRE CA trust bundle. No pre-shared secrets, no sidecar injection of credentials, no service-specific token. If the signature checks out and the SPIFFE ID matches an access policy, the connection is accepted.

Steps 0–3 are handled entirely by SPIRE. Steps 4–5 are between the workload and whatever it is talking to; SPIRE's job is to get a cryptographically-grounded identity into the workload's hands without the workload needing to manage secrets itself.

Node Attestation

The agent can't just claim "I'm a legitimate node" — the server needs proof. **Node attestation** is how the agent proves to the server that it's running on a real, authorized Kubernetes node.

This cluster uses **k8s_psat** (Kubernetes Projected Service Account Token):

1. Kubernetes issues a short-lived projected SAT to the agent pod (audience: `spire-server`, expiry: 7200s). This token is in the pod's filesystem at `/var/run/secrets/tokens/`.
2. The agent sends this token to the SPIRE server over gRPC.
3. The server calls the Kubernetes **TokenReview API** to validate it — Kubernetes confirms the token is real, which SA issued it, and which pod it was issued to.
4. The server checks its allow-list (`spire:spire-agent SA`) and grants the agent a **Node SVID** with SPIFFE ID `spiffe://ipc.local/k8s-node`.

From that point on, the agent authenticates to the server using the Node SVID (mTLS on the gRPC channel). The PSAT is only used once for initial bootstrapping.

Why PSAT instead of something else? These nodes have no TPM, no cloud provider identity (AWS EC2 instance identity documents, GCP GCE tokens, etc.), and no pre-provisioned join token — PSAT is the right choice for bare-metal Kubernetes. The trust root is “Kubernetes itself validated this token.”

Note: this cluster has been upgraded to TPM DevID node attestation (`tpm_devid`). The `k8s_psat` description above is retained as conceptual background; the actual implementation uses the hardware TPM and is documented in the TPM Attestation sections.

TPM DevID Node Attestation Flow

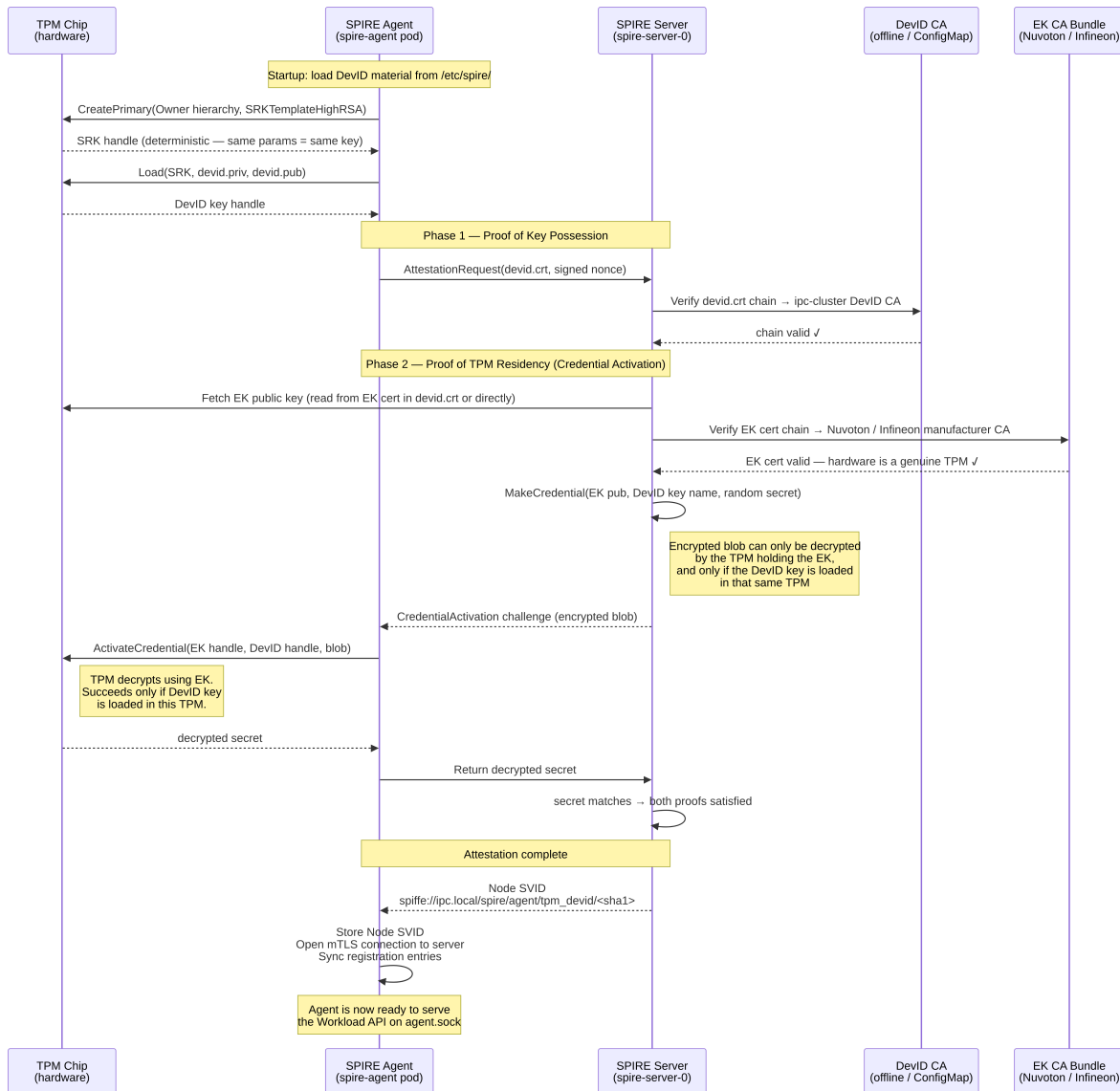


Figure 2: TPM DevID node attestation sequence

Workload Attestation

After a node is attested, the agent needs to verify that a calling workload is what it claims to be before issuing an SVID. This happens entirely on-node; no server round-trip is needed. Critically, **the workload never gets to assert its own identity** — the agent determines it entirely by observation from outside the workload process.

The flow for this cluster (k8s workload attestor):

1. Workload connects to the agent socket. The workload process opens a connection to `/run/spire/sockets/agent.sock`, a Unix domain socket on the node filesystem, mounted into the workload pod via a `hostPath` volume.

2. Agent reads `SO_PEERCRE`D → host PID. Unix domain sockets have a kernel feature called `SO_PEERCRE`D. When the agent calls `getsockopt` with that option on the connected socket, the kernel returns the PID, UID, and GID of the process on the other end. This is trustworthy because it comes from the kernel — the connecting process cannot forge it. The result is the *host* PID (e.g. 48721), which is why `hostPID: true` is required on the agent `DaemonSet`: without it, PIDs are namespaced inside the pod and `SO_PEERCRE`D returns a PID that is meaningless outside that namespace.

3. Agent reads `/proc/<pid>/cgroup` → container ID. The cgroup path for a containerized process encodes the container ID, in a form like:

```
0::/kubepods/besteffort/pod<pod-uid>/<container-id>
```

The agent parses this to extract the container ID. (`use_new_container_locator: true` in the agent config tries `mountinfo` first, which is more reliable under `cgroupv2`, before falling back to the `cgroup` path.)

4. Agent calls kubelet API → pod metadata. The agent calls the local kubelet’s `/pods` endpoint (this is why the agent needs `nodes/proxy` RBAC) asking: “which pod contains a container with this ID?” The kubelet returns the full pod record: namespace, pod name, service account, labels, and annotations. The agent now has everything Kubernetes knows about who made the connection.

5. Agent matches pod metadata against registration entries. The agent compares the pod’s namespace and service account against its registration entries (synced from the SPIRE server). If an entry matches — for example, `ns=mtls-demo, sa=mtls-client-sa → spiffe://ipc.local/mtls-client` — the agent issues an SVID for that SPIFFE ID.

The chain is: *which process connected* (kernel) → *which container holds that process* (procfs) → *which pod owns that container* (kubelet) → *which SPIFFE ID applies* (registration entries). Each step is derived from a source the workload cannot influence.

Pelagos Compatibility Fixes

Two Pelagos bugs affected SPIRE workload attestation on this cluster:

Bug	Symptom	Fixed
<code>hostPID: true</code> ignored (#299)	<code>SO_PEERCRE</code> D returned PID 0; server rejected	v0.65.7

Bug	Symptom	Fixed
32-char container IDs (#301)	SPIRE regex expected 64-char hex; attestation failed	v0.65.8

Both are fixed in the running version (v0.65.47).

Workload Attestation Flow

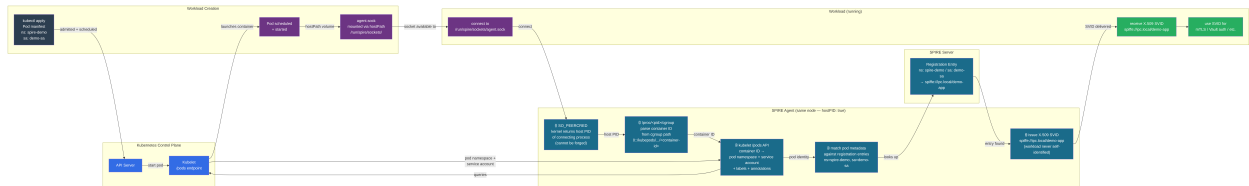


Figure 3: Workload attestation flow

Registration Entries

Registration entries are the policy database — they tell the server (and thus agents) which workloads should receive which SPIFFE IDs. Each entry has:

- **SPIFFE ID:** the identity to issue (`spiffe://ipc.local/demo-app`)
- **Parent ID:** who must attest the caller first (`spiffe://ipc.local/k8s-node` — meaning “the workload must be on an attested node”)
- **Selectors:** the conditions that must all match (`k8s:ns:spire-demo, k8s:sa:demo-sa`)

Current Pattern: Imperative Registration

A Kubernetes Job runs `kubectl exec` into the server pod and calls `spire-server entry create`. This works but has a GitOps gap — entries live in SQLite on the NFS PVC, not in Git. If the PVC is lost or the cluster is rebuilt, entries must be re-registered.

GitOps Alternative: ClusterSPIFFEID

The **SPIRE Controller Manager** provides a `ClusterSPIFFEID` CRD. Entries are declared as Kubernetes resources in Git; the controller reconciles them into SPIRE continuously. This closes the GitOps gap. Architected in `docs/spire-trust-chain-clusterspiffeid.mmd` but not yet deployed.

SVIDs — the Two Forms

SVID = SPIFFE Verifiable Identity Document. Two formats:

X.509 SVID

A standard X.509 certificate with one key addition: the SPIFFE ID is encoded in the certificate's **Subject Alternative Name (SAN)** as a URI (`spiffe://ipc.local/demo-app`). Inspect with:

```
openssl x509 -in svid.pem -text -noout | grep URI
```

Properties:

- **Short-lived** (default TTL: 1 hour). The agent rotates them automatically before expiry, typically at the 50% mark.
- Carries the full chain: leaf cert + intermediate(s) up to the trust bundle.
- Used for **mTLS** — both sides present their SVID as the TLS client/server cert, and verify the peer's cert against the trust bundle. This is what experiment 21 demonstrates.
- Opaque to network intermediaries (the identity is in the cert, not the payload).

JWT SVID

A signed JWT with the SPIFFE ID in the `sub` claim and an `aud` (audience) claim.

Properties:

- Also short-lived. **Not** automatically rotated — the workload must re-fetch when it needs a fresh one.
- Used when mTLS is not possible: HTTP APIs where identity must be passed in a header (`Authorization: Bearer <jwt>`), or when a network intermediary (load balancer, API gateway) terminates TLS before reaching the workload.
- The receiving service verifies the JWT signature against the trust bundle's public key, not by doing a TLS handshake.
- Fetch with: `spire-agent api fetch jwt -audience <target-service> -socketPath ...`
- The **audience field matters**: a JWT fetched for audience `api-gateway` won't be accepted by a verifier expecting `payments-service`. This prevents token replay across services.

When to Use Which

Scenario	Format
Service-to-service where you control both ends	X.509 (mTLS)
HTTP API, identity passed in header	JWT
Identity through a TLS-terminating proxy	JWT
Authenticating to Vault (JWT auth method)	JWT
Envoy/service mesh transparent mTLS	X.509

The Trust Bundle

The trust bundle is the SPIRE CA's root certificate(s). Everything signed by SPIRE is verifiable against it. On this cluster:

- The server writes it to the `spire-bundle` ConfigMap in the `spire` namespace (via the `k8s_bundle` notifier).
- Agents mount it at startup to bootstrap their TLS connection to the server.
- Workloads receive it alongside their SVID when calling the Workload API — so they can verify peers without any pre-shared secrets or manual cert distribution.

Trust bundle rotation: If the server’s CA rotates (either on schedule or forced), it publishes the new root alongside the old one during a transition window. SVIDs issued under the old CA remain valid until expiry; verifiers must accept both roots during the overlap. This is automatic in SPIRE.

Experiments Running on This Cluster

Experiment 11: Workload SVID Fetch

Namespace `spire-demo`, service account `demo-sa`. A demo pod fetches its X.509 SVID via the Workload API and prints the SPIFFE ID and cert details using `openssl`. Proves end-to-end attestation works.

Registration entries created by an imperative Job:

SPIFFE ID	Parent	Selectors
<code>spiffe://ipc.local/k8s-node</code>	<code>spiffe://ipc.local/spire-server</code>	<code>k8s:server:cluster:ipc</code> (node entry)
<code>spiffe://ipc.local/demo-app</code>	<code>spiffe://ipc.local/k8s-k8s:ns:spire-demo,node</code>	<code>k8s:ns:spire-demo,k8s:sa:demo-sa</code>

Experiment 21: mTLS Between Two Workloads

Namespace `mtls-demo`. An `mtls-server` and `mtls-client` each receive unique X.509 SVIDs from SPIRE. The server runs `openssl s_server` requiring client cert (`-Verify 1`); the client runs `openssl s_client` presenting its SVID. Both sides verify against the SPIRE trust bundle. Zero pre-shared secrets.

SPIFFE ID	Workload
<code>spiffe://ipc.local/mtls-server</code>	<code>mtls-server</code> Deployment (<code>mtls-server-sa</code>)
<code>spiffe://ipc.local/mtls-client</code>	<code>mtls-client</code> Pod (<code>mtls-client-sa</code>)

TPM Hardware on This Cluster

All six nodes have TPM 2.0 chips with manufacturer-provisioned Endorsement Key (EK) certificates stored in TPM NV (non-volatile) memory. The EK cert is the hardware root of trust: it is signed by the manufacturer’s CA at the factory and proves that the EK private key is bound to a specific, real TPM chip. The EK private key never leaves the chip.

EK Certificate Inventory

Node	Chip	Manufacturer	EK Cert Issuer	Valid Until
ipc4	Nuvoton NPCT75x	Nuvoton (TW)	NPCTxxx ECC384 LeafCA 022111	2043
ipc5	Nuvoton NPCT75x	Nuvoton (TW)	NPCTxxx ECC384 LeafCA 022111	2043
ipc6	Nuvoton NPCT75x	Nuvoton (TW)	NuvotonTPM- RootCA2210	2042
ipc7	Nuvoton NPCT75x	Nuvoton (TW)	NuvotonTPM- RootCA2211	2042
ipc8	Infineon SLB9672	Infineon (DE)	OPTIGA(TM) TPM 2.0 RSA CA 061	2037
ipc9	Infineon SLB9672	Infineon (DE)	OPTIGA(TM) TPM 2.0 RSA CA 061	2037

`tpm2-tools` is installed on all nodes (`apt install tpm2-tools`). The EK cert can be read with `sudo tpm2_getekcertificate -o /tmp/ek.crt`.

Are Nuvoton and Infineon Equivalent?

For SPIRE attestation: yes, completely. The TPM 2.0 attestation protocol is standardized by the TCG (Trusted Computing Group) and is identical regardless of manufacturer. SPIRE validates the EK cert chain and does a challenge-response over the EK private key; the manufacturer is irrelevant to the protocol.

In general terms, Infineon OPTIGA is the dominant TPM manufacturer globally and is used in FIPS-certified and government-cleared systems. Nuvoton NPCT is a legitimate TCG-certified TPM 2.0 part commonly shipped by HP in their business machines. Neither has active unpatched vulnerabilities. The historical ROCA vulnerability (CVE-2017-15361) affected Infineon RSA key *generation* in old firmware; ipc8/9 run firmware 15.0.0.22, well past the patched version.

ipc8/9 have both RSA and ECC EK certs (Infineon provisions both; Nuvoton nodes only have one). ECC P-384 is more modern than RSA 2048 — a mild theoretical advantage, not meaningful in practice.

ipc6 and ipc7 have different Nuvoton root CAs (2210 vs 2211) despite being the same hardware model. This reflects different manufacturing batches. For SPIRE attestor configuration you will need to trust all four manufacturer CA chains: two Nuvoton, one Infineon RSA, one Infineon ECC.

What Happens After 2037 (ipc8/9 EK Cert Expiry)?

You do not replace the TPM or the hardware. The TPM chip is permanent — it's soldered to the motherboard. What expires in 2037 is the manufacturer-signed *certificate* for the EK public key, not the EK key itself. The hardware continues to function normally.

What changes is your attestation policy. Currently, full EK cert chain validation works like this: “I trust this node because its EK cert chains to Infineon’s CA, which means Infineon vouches that this is a real TPM.” After expiry, that chain is stale.

The alternative — which is always available and is what most production deployments use alongside cert-chain validation — is **EK public key fingerprinting**: during initial enrollment, you record the EK public key of each node. The attester then checks: “I trust this node because it can prove possession of the private key corresponding to *this enrolled public key*.” There is no expiry on a public key fingerprint. You lose the manufacturer-provenance guarantee (that the key is in a real TPM) but retain the cryptographic identity guarantee (that this is the same hardware you enrolled).

In practice for this cluster: re-enrollment in 2037 is the right move — you run through node attestation setup once more at that point and record the EK fingerprints. Or you pre-enroll fingerprints now alongside cert chain validation, and the transition in 2037 is a config change rather than a re-enrollment.

TPM Attestation: Plugin Choice and Implementation Plan

Plugin: `tpm_devid` (built-in to SPIRE)

SPIRE 1.9.6 has a native TPM node attester called `tpm_devid`, stable since SPIRE 1.0.1 (2021). No external plugins are needed. The Bloomberg community plugin (`bloomberg/spire-tpm-plugin`) is archived and unmaintained — ignore it.

How `tpm_devid` Attestation Works

The attestation is two-part, which is what makes it stronger than a simple hardware value check:

1. **Proof of key possession** — the agent signs a server-issued nonce with its DevID private key. The server verifies the signature against the DevID certificate’s public key.
2. **Proof of TPM residency** — the server performs a **credential activation challenge** using the node’s EK public key. It encrypts a secret to the EK such that only the TPM holding the EK can decrypt it, and only if the DevID key is also loaded in that same TPM. The agent returns the decrypted secret, proving both keys reside in the same physical chip.

This two-part proof means an attacker can’t just copy key material off a node — possession of the DevID private key blob is not enough; it must be activated inside the TPM that holds the EK.

The SPIFFE ID issued by the `tpm_devid` attester takes the form: `spiffe://ipc.local/spire/agent/tpm_devid/<fingerprint-of-DevID-cert>`

What `tpm_devid` Requires

Per node (out-of-band provisioning): - A **DevID key pair generated inside the TPM** — the private key never leaves the chip; it exists only as an encrypted TPM key blob (`devid.priv`) paired with a public key blob (`devid.pub`) - A **DevID certificate** signed by a CA you control (the DevID CA), containing the public key

SPIRE Server: - `devid_ca_path` — the DevID CA cert (validates the DevID cert chain on each agent) - `endorsement_ca_path` — all manufacturer EK CA certs (validates the EK cert chain, completing the hardware root-of-trust verification)

This cluster requires four EK CA certs: Nuvoton NPCTxxx LeafCA 022111, NuvotonTPM-RootCA2210, NuvotonTPMRootCA2211, and Infineon OPTIGA TPM 2.0 RSA CA 061.

SPIRE Agent: - `devid_cert_path` — path to the signed DevID certificate (PEM) - `devid_priv_path` — path to the TPM private key blob - `devid_pub_path` — path to the TPM public key blob

Migration Impact

Switching from `k8s_psat` to `tpm_devid` invalidates all existing agent registrations — the Node SVID format changes. The SPIRE server's registration database must be cleared and the registration jobs re-run after the switch. Since entries are currently managed by imperative Jobs, this is re-running those jobs, not a data migration.

TPM Attestation: Implementation — What Was Built

All seven phases are complete. This section documents what was actually implemented, including the non-obvious technical details discovered during the process.

Phase 1 — DevID CA (complete)

An offline EC P-256 CA (20-year validity) was generated on omen and used exclusively to sign node DevID certificates.

- **Private key:** stored in 1Password as `ipc-cluster DevID CA key` (Private vault, notes field). Never written to disk longer than the signing operation; shredded immediately after storage.
- **Public cert:** `config/tpm/devid-ca.pem` — committed to Git (public material only).
- **Signing:** the provisioning script retrieves the key via `op item get ... | tr -d '"'` (the `op` CLI wraps field values in double quotes; stripping is required for the PEM to be valid).

Phase 2 — Manufacturer EK CA Bundle (complete)

Six CA certificates assembled into `config/tpm/endorsement-ca-bundle.pem`. All URLs were discovered from the AIA (Authority Information Access) extensions in the live EK certificates on each node via `tpm2_getekcertificate + openssl x509 -text`.

The bundle is two full chains, not four flat certs:

Chain	Nodes	Certs
Nuvoton ECC521	ipc4, ipc5	Root (NPCTxxx ECC521 RootCA) → Intermediate (NPCTxxx ECC384 LeafCA 022111)
Nuvoton 2210	ipc6	Self-signed root (NuvotonTPMRootCA2210)
Nuvoton 2211	ipc7	Self-signed root (NuvotonTPMRootCA2211)
Infineon RSA	ipc8, ipc9	Root (OPTIGA RSA Root CA 2) → Intermediate (OPTIGA TPM 2.0 RSA CA 061)

The EK CA bundle is the proof that an EK cert was signed by a real manufacturer. Without it, SPIRE cannot verify that the TPM presenting the EK is a genuine hardware chip.

Phase 3 — Node Provisioning Script (complete)

`scripts/provision-tpm-devid.sh` runs from `omen`, provisions one node end-to-end, and is idempotent (skips if a valid cert signed by our CA already exists with >30 days to expiry).

What the script does:

1. Installs `tpm2-openssl` on the node (`apt install tpm2-openssl`)
2. Creates the SRK (Storage Root Key) primary under the **Owner hierarchy** (`-C o`)
3. Creates the DevID child key under the SRK
4. Loads the key and persists it to a temporary NV handle (`0x81010001`)
5. Generates a CSR via the `tpm2` OpenSSL provider — the TPM signs the CSR internally
6. Evicts the NV handle — SPIRE uses blob files, not handles
7. Strips the 2-byte TPM2B wrapper from both blob files (see below)
8. Scp's the CSR to `omen`; retrieves the DevID CA key from `1Password`; signs the cert
9. Installs the signed cert at `/etc/spire/devid.crt`; verifies it chains to the DevID CA
10. Cleans up temporary files; leaves `/etc/spire/` with `devid.crt`, `devid.priv`, `devid.pub`, `devid_pubkey.pem`

Critical implementation details discovered during integration:

Primary key hierarchy: The primary must be under the **Owner hierarchy** (`-C o`), not the Endorsement hierarchy (`-C e`). SPIRE's `tpm_devid` plugin recreates the primary at startup using `SRKTemplateHighRSA()` from `go-tpm-tools`, which calls `tpm2.CreatePrimaryEx` against `tpm2.HandleOwner`. In the TPM, primary keys are deterministic — same parameters + same hierarchy = same key. If the provisioning script uses the Endorsement hierarchy but SPIRE uses the Owner hierarchy, the primaries are different keys and SPIRE cannot load the DevID child blobs.

Primary key attributes (must match SPIRE's `SRKTemplateHighRSA()` exactly):

```
-G rsa
-a "fixedtpm|fixedparent|sensitiveorigin|userwithauth|noda|restricted|decrypt"
```

The symmetric cipher (AES-128-CFB) is implied by `restricted|decrypt` in `tpm2-tools` 5.7. Do NOT use `-G rsa2048:aes128cfb:null` — that compound form is not accepted.

DevID key signing scheme must be `rsa:rsassa` (RSASSA-PKCS1-v1.5), not `rsa` (which defaults to `null` scheme). With a `null` scheme, `go-tpm`'s `DecodePublic` returns a `Public` struct where `RSAPParameters.Sign` is `nil`, causing a `nil` pointer dereference in the `tpm_devid` plugin.

Blob format — stripping the TPM2B prefix: `tpm2_create -u/-r` outputs blobs in `TPM2B_PUBLIC` / `TPM2B_PRIVATE` format (2-byte big-endian size + payload). SPIRE's `go-tpm` library expects the raw payload without that prefix: `- tpm2.DecodePublic(pubBytes)` decodes raw `TPMT_PUBLIC` directly - `tpm2.Load(... privateBlob)` packs the private blob via `U16Bytes()`, adding the prefix itself — so the file must NOT already have one

The strip must happen **after** `tpm2_load` (which needs the full TPM2B wrappers to load the key), not before:

```
dd if=devid.pub bs=1 skip=2 of=devid.pub.raw && mv devid.pub.raw devid.pub
dd if=devid.priv bs=1 skip=2 of=devid.priv.raw && mv devid.priv.raw devid.priv
```

CSR generation via tpm2-openssl: Standard openssl req -key cannot use a TPM public key PEM as if it were a private key. Instead: install tpm2-openssl, persist the DevID key to a TPM NV handle, and generate the CSR through the TPM provider:

```
openssl req -provider tpm2 -provider default -key "handle:0x81010001" -new -out devid.csr ...
```

The private key signs the CSR data inside the TPM chip. After the CSR is fetched, evict the handle — SPIRE only needs the blob files.

Phase 4 — SPIRE Server Config (complete)

manifests/spire/server-config.yaml: k8s_psat attestor replaced with tpm_devid:

```
NodeAttestor "tpm_devid" {
  plugin_data {
    devid_ca_path      = "/run/spire/tpm-ca/devid-ca.pem"
    endorsement_ca_path = "/run/spire/tpm-ca/endorsement-ca-bundle.pem"
  }
}
```

manifests/spire/server-tpm-ca.yaml: new ConfigMap carrying both CA files, mounted into the StatefulSet at /run/spire/tpm-ca/.

manifests/spire/server-rbac.yaml: the server ClusterRole was stripped of tokenreviews, subjectaccessreviews, nodes, and pods. With tpm_devid, the server does not call the Kubernetes API at all — attestation is purely TPM-driven. The Role (managing the spire-bundle ConfigMap) was kept.

Phase 5 — SPIRE Agent Config and DaemonSet (complete)

manifests/spire/agent-config.yaml: k8s_psat replaced with tpm_devid:

```
NodeAttestor "tpm_devid" {
  plugin_data {
    devid_cert_path = "/etc/spire/devid.crt"
    devid_priv_path = "/etc/spire/devid.priv"
    devid_pub_path  = "/etc/spire/devid.pub"
    tpm_device_path = "/dev/tpmrm0"
  }
}
```

tpm_device_path is set explicitly. Without it, the plugin auto-detects, which failed inside the container even after the device was mounted.

manifests/spire/agent-daemonset.yaml: two volume changes: - **Added:** /dev/tpmrm0 as a CharDevice hostPath — the SPIRE agent must open the TPM device directly to execute the Load command and credential activation challenge. /dev/tpmrm0 (the resource manager) is used rather than /dev/tpm0 (the raw device) because the resource manager handles concurrent access from multiple processes. - **Removed:** projected service account token volume (spire-token) — no longer needed since node attestation is TPM-based, not PSAT-based.

Phase 6 — Deploy and Re-register (complete, 2026-07-09)

Server rolled first. Log confirms:

Plugin loaded: plugin_name=tpm_devid plugin_type=NodeAttestor

All 6 agents attested successfully. Each node received a unique SPIFFE ID derived from the SHA-1 fingerprint of its DevID certificate:

Node	SPIFFE ID (agent)
ipc4	spiffe://ipc.local/spire/agent/tpm_devid/25a1f67bfc2a6dc830de61d8dd2b5e78
ipc5	spiffe://ipc.local/spire/agent/tpm_devid/37bec218b1f39af8e7db42f04d710a3a
ipc6	spiffe://ipc.local/spire/agent/tpm_devid/ab96c7507b860042e3711446552e8d2e
ipc7	spiffe://ipc.local/spire/agent/tpm_devid/8b84df1549516ddba24415505b989fbb
ipc8	spiffe://ipc.local/spire/agent/tpm_devid/27bd47f544c71bd8ed20ec07c546a225
ipc9	spiffe://ipc.local/spire/agent/tpm_devid/dd04bfbf9dbfb604b0fdc1bf1d42d03e

Registration entries updated: a new node alias entry was added alongside the existing k8s_psat entry:

SPIFFE ID: spiffe://ipc.local/k8s-node

Parent: spiffe://ipc.local/spire/server

Selector: tpm_devid:issuer:cn:ipc-cluster DevID CA

-node flag set

The issuer:cn selector matches all nodes signed by our DevID CA without needing per-node entries. Workload entries (demo-app, mtls-server, etc.) were unchanged — they reference spiffe://ipc.local/k8s-node as their parent, which the new alias entry satisfies.

End-to-end verification: the spire-demo/demo-workload pod received SVID spiffe://ipc.local/demo-app (1-hour TTL, signed by the SPIRE CA).

Phase 7 — Reinstall Automation (complete)

scripts/reinstall-nodes.sh calls provision-tpm-devid.sh after a node rejoins k3s and is Ready. If op is not authenticated, the step warns but does not fail the reinstall. The operator can re-run bash scripts/provision-tpm-devid.sh <node> manually afterward.

Operational loop for a reinstalled node: 1. PXE reinstall completes 2. Node rejoins k3s (join-server.sh or upgrade-agents.sh) 3. provision-tpm-devid.sh generates a new TPM-bound DevID key, fetches CA key from 1Password, signs a new cert, installs at /etc/spire/ 4. Flux reconciles SPIRE agent DaemonSet — agent starts, loads DevID blobs, attests via TPM 5. SPIRE server issues Node SVID matching tpm_devid:issuer:cn:ipc-cluster DevID CA 6. Workloads on the node can receive SVIDs

No manual SPIRE steps are needed after a node reinstall.

Topics Not Asked About That Matter

SVID Rotation for Long-Running Workloads

The experiments use an **init-container pattern**: fetch SVID once at startup, write to an emptyDir, main container uses it. This is fine for demos but won't handle rotation — after ~1 hour the cert expires and the workload has stale credentials.

For production workloads, the options are:

- **spiffe-helper**: a sidecar that watches the Workload API socket for updated SVIDs and rewrites cert files on disk, then sends SIGHUP to the main process.
- **SPIFFE SDK** (Go, Java, Python): native library that holds a live connection to the Workload API and delivers updated SVIDs to the application via callback.
- **Envoy SDS** (Secret Discovery Service): Envoy natively speaks the SPIFFE Workload API; it handles rotation transparently without application changes.

SPIRE HA

The current server is a single StatefulSet replica with SQLite. SQLite cannot be shared across replicas. High availability requires:

1. Switch the datastore to PostgreSQL or MySQL
2. Run multiple server replicas behind a load balancer

With a single server, a server restart means agents can't renew SVIDs during the outage. Workloads continue with cached SVIDs until they expire. Acceptable for a homelab; a consideration for production.

Federation

SPIRE supports federating trust between different SPIFFE trust domains. For example, `ipc.local` and `cloud.example.com` could exchange trust bundles so workloads in each cluster can verify SVIDs from the other. Not needed until there are multiple clusters or external partners, but the mechanism is built in to SPIRE.

The Trust Root and Security Boundary

With `tpm_devid`, the server's trust in the agent is hardware-rooted: the agent proves possession of a TPM-bound key and that the key and EK live in the same physical chip. This is strong, asymmetric cryptographic proof.

The agent's trust in the server is not symmetric. It is Kubernetes-rooted.

The bootstrap problem: before the agent can attest, it needs to establish a TLS connection to the SPIRE server. To verify the server's TLS certificate it needs a trust anchor — the SPIRE CA cert — which arrives in the `spire-bundle` ConfigMap. But the ConfigMap is not itself cryptographically authenticated. The agent has no independent way to verify “this CA cert belongs to *the* SPIRE server I am supposed to talk to.”

What the agent actually knows at bootstrap:

1. There is a CA cert at `/run/spire/bundle/bundle.crt`
2. The server it can reach presents a TLS cert signed by that CA

3. Those two facts are mutually consistent

Consistency is not proof of legitimacy. A rogue SPIRE server could publish its own CA cert into the ConfigMap and present a cert signed by it. From the agent’s perspective, both situations look identical — cert matches CA, TLS handshake succeeds, proceed to attest. The agent cannot detect the substitution.

The circularity: the agent uses the bundle to authenticate the server, but the bundle itself arrived from an unauthenticated channel. Breaking this circularity requires an out-of-band trust anchor — something the agent knows before it ever contacts the server.

Current posture: the ConfigMap write is protected by RBAC. Only the `spire-server` ServiceAccount can write `spire-bundle`. This is a policy control, not a cryptographic proof. If that policy is ever violated — through a compromised ServiceAccount, a misconfigured RBAC rule, or a cluster-admin mistake — the agent cannot detect it. The `spire` namespace is the security boundary, not individual workloads.

TPM-Based Server Attestation (the fix — implemented 2026-07-09)

The `spiffe/spire-server-attestor-tpm` project (v0.0.4, April 2025) closes this gap. It provides four binaries that together give agents a cryptographic way to verify the server before accepting its bundle.

Components

Binary	Role	Runs on
<code>spire-server-attestor-tpm-sign</code>	SPIRE <code>BundlePublisher</code> plugin; receives the trust bundle from the server and pushes it to the signer	SPIRE server pod (hostPath binary)
<code>spire-server-attestor-tpm-signer-unix</code>	Root daemon with TPM access; signs the bundle using the server’s TPM key; writes signed JWT to disk	SPIRE server pod (hostPath binary, <code>/dev/tpmrm0</code> mounted)

Binary	Role	Runs on
<code>spire-server-attestor-tpm-signer-http</code>	Not used on this cluster. Despite its name, this is a software-key signing alternative to <code>signer-unix</code> — it signs JWTs using a PEM RSA private key, not the TPM. It is not an HTTP frontend for <code>signer-unix</code> . Using it here would replace hardware attestation with a software key.	—
<code>spire-server-attestor-tpm-verifier</code>	Not used on this cluster. Agent-side daemon intended for systemd deployments; replaced here by the <code>verify-bundle</code> init container.	—

The project ships all four binaries. Two are appropriate for this cluster’s Kubernetes/TPM deployment; two are not:

- **sign + signer-unix:** use these. They provide the TPM-backed signing chain.
- **signer-http:** skip. It is for deployments *without* a TPM, where a software key is acceptable. Deploying it alongside `signer-unix` would serve a separately signed (software-keyed) bundle that agents could not verify against the TPM public key anyway.
- **verifier:** skip. Designed as a persistent daemon for systemd-managed agents. Replaced here by the `verify-bundle` init container, which is better suited to the Kubernetes pod lifecycle (runs once, exits, agent starts).

The HTTP serving role that `signer-http` would fill in a non-TPM deployment is handled here by a `busybox httpd` sidecar — one line, no config, serves the file that `signer-unix` writes.

Data Flow

Why This Breaks the Circular Trust The verifier must verify the HTTP bundle before the SPIRE agent accepts it. The verification requires the server’s TPM public key. That public key is NOT fetched from Kubernetes — it is written to `/etc/spire/server-bundle-signing.pub` on each agent node during physical provisioning (by `scripts/provision-tpm-devid.sh`), before any SPIRE component starts.

This is the out-of-band anchor. An attacker who compromises the Kubernetes control plane and rewrites the `spire-bundle` ConfigMap or the `signer-http` response still cannot produce a valid signature over the substituted bundle because:

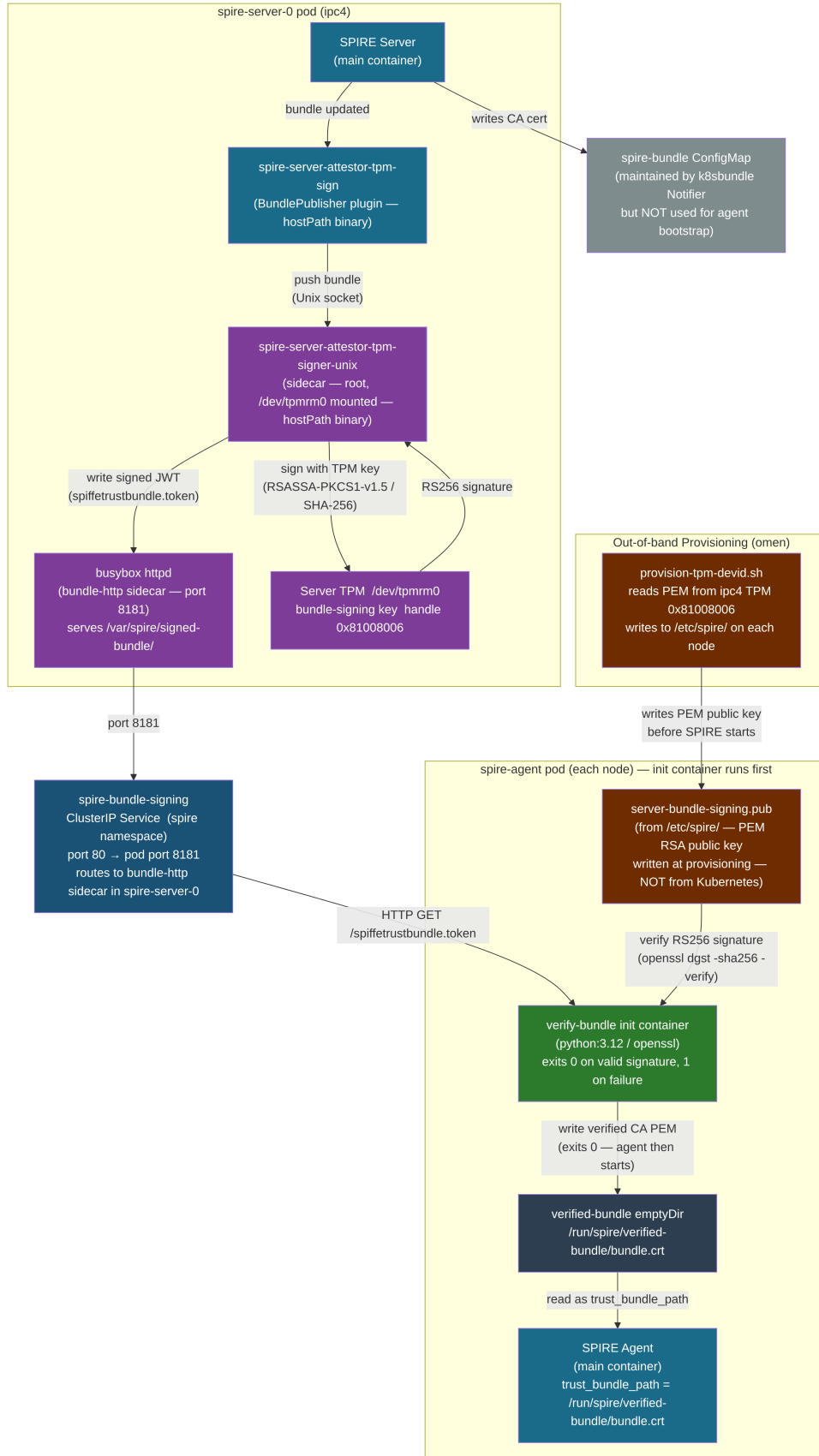


Figure 4: TPM server bundle signing and verification flow

- Producing the signature requires the server's TPM private key
- That private key never leaves ipc4's TPM chip
- The verifier checks the signature before the SPIRE agent accepts anything
- Signature check fails → agent rejects the bundle → does not connect to the rogue server

The server's TPM public key on the agent nodes is the cryptographic root of the entire bootstrap chain. It is distributed through the same physical provisioning channel as the agent's own DevID material, not through Kubernetes.

Server TPM Key (separate from DevID) A dedicated **bundle-signing key** is provisioned on ipc4's TPM, separate from the agent's DevID key. Key separation is important: the DevID key is used for SPIRE agent attestation; the bundle-signing key is used only to authenticate the trust bundle to agents. Different purposes, different keys, both TPM-bound.

The bundle-signing key uses TPM handle 0x81008006 (the handle recommended by the project). It is created under the Owner hierarchy (same as DevID keys) with RSA 2048 and `rsassa` signing scheme.

Integration for This Cluster Because our SPIRE server and agents run as Kubernetes pods rather than systemd services, the components are adapted as pod sidecars:

- `signer-unix` and `signer-http` run as sidecar containers in the `spire-server` StatefulSet pod. `signer-unix` has `/dev/tpmrm0` mounted.
- The `sign` plugin binary is a `hostPath` volume mounted from `/usr/local/bin/spire-server-attestor-tpm-sign` on ipc4 (the server node). Similarly, the `signer-unix` binary is mounted from `/usr/local/bin/spire-server-attestor-tpm-signer-unix`.
- The HTTP endpoint is exposed as a ClusterIP Kubernetes Service.
- Instead of the `verifier` binary, a **verify-bundle init container** (`docker.io/library/python:3.12`) runs the verification at agent pod startup. It fetches the signed JWT from the `spire-bundle-signing` ClusterIP Service (the `nginx` sidecar), verifies the RS256 signature using `openssl dgst -sha256 -verify` against `/etc/spire/server-bundle-signing.pub`, extracts the `spiffetb` payload, and writes it to `/run/spire/verified-bundle/bundle.crt` on a shared `emptyDir` volume. The agent pod does not start until this init container exits 0.
- The SPIRE agent uses `trust_bundle_path = "/run/spire/verified-bundle/bundle.crt"` (the `emptyDir` written by the init container) instead of the unauthenticated `spire-bundle` ConfigMap.
- The `bundle-http` sidecar in the server pod is **`nginx:alpine`** (not the dedicated `signer-http` binary from the project). `nginx` serves `/var/spire/signed-bundle/` on port 8181; the ConfigMap `spire-signer-unix-config` contains both the `signer-unix` config and an `nginx` `vhost` config with `Cache-Control: no-store`.
- `scripts/provision-tpm-devid.sh` is extended to also write `/etc/spire/server-bundle-signing.pub` on every agent node.

Resulting Security Posture

Direction	Before (k8s_psat era)	After tpm_devid	After server attestation
Server trusts agent	Kubernetes TokenReview (policy)	TPM credential activation (hardware)	TPM credential activation (hardware)
Agent trusts server	ConfigMap (unauthenticated)	ConfigMap (unauthenticated)	TPM signature on bundle (hardware)
Attacker must compromise	RBAC policy	Physical TPM on agent node	Physical TPM on both ipc4 AND agent node
Can agent detect rogue server?	No	No	Yes — signature check fails

Vault Integration via JWT SVID

A natural next step for this cluster (Vault is already deployed): pods can authenticate to Vault using a SPIRE-issued JWT SVID via Vault's JWT auth method. No Vault tokens or secrets need to be injected into pods. The flow is:

1. Pod fetches JWT SVID (audience: `vault`)
2. Pod calls `vault write auth/jwt/login jwt=<svid> role=<role>`
3. Vault validates the JWT against the SPIRE trust bundle (configured as a JWKS endpoint)
4. Vault returns a short-lived Vault token

This replaces Vault's Kubernetes auth method with a stronger, SPIFFE-native identity.

Configuration Reference

Key Files

Path	Purpose
<code>manifests/spire/server-config.yaml</code>	SPIRE Server ConfigMap (trust domain, attestors, storage)
<code>manifests/spire/agent-config.yaml</code>	SPIRE Agent ConfigMap (server address, attestors, socket path)
<code>manifests/spire/server-statefulset.yaml</code>	Server StatefulSet + headless service
<code>manifests/spire/agent-daemonset.yaml</code>	Agent DaemonSet (hostPID, tolerations, socket hostPath)
<code>manifests/spire/server-rbac.yaml</code>	Server SA, ClusterRole (TokenReview, nodes, pods)
<code>manifests/spire/agent-rbac.yaml</code>	Agent SA, ClusterRole (pods, nodes, nodes/proxy)
<code>clusters/ipc/spire.yaml</code>	Flux Kustomization (path: <code>./manifests/spire</code> , interval: 10m)
<code>experiments/11-spire/</code>	Workload SVID fetch demo
<code>experiments/21-mtls-spire/</code>	mTLS between two workloads

Server Configuration Summary

Setting	Value
Trust domain	<code>ipc.local</code>
Node attester	<code>tpm_devid; DevID CA: config/tpm/devid-ca.pem; EK CA bundle: config/tpm/endorsement-ca-bundle.pem</code>
Key manager	Disk (<code>/run/spire/data/keys.json</code>)
Datastore	SQLite3 (<code>/run/spire/data/datastore.sqlite3</code>)
Notifier	<code>k8s_bundle</code> → ConfigMap <code>spire-bundle</code>
BundlePublisher	<code>tpm-sign</code> plugin → signer-unix socket; frequency 5m
gRPC port	8081
Health port	8080 (<code>/live</code> , <code>/ready</code>)

Agent Configuration Summary

Setting	Value
Trust domain	<code>ipc.local</code>
Server	<code>spire-server:8081</code>
Trust bundle	<code>/run/spire/verified-bundle/bundle.crt</code> (emptyDir written by <code>verify-bundle</code> init container)
Workload API socket	<code>/run/spire/sockets/agent.sock</code>
Node attester	<code>tpm_devid; DevID blobs from /etc/spire/; /dev/tpmrm0</code>
Workload attester	<code>k8s, skip_kubelet_verification: true, use_new_container_locator: true</code>
Key manager	Memory
Health port	8080